Silverio Martínez Andrés Grupo 36 Tarea 18

Nombre:
Tema:
Día Mes Año Folio

Cal. 10

Tarea 18

1) Verificar que:

a) $\lim_{x\to 0} \frac{tg\,x}{x} = 1$

$\frac{tg\,0}{0} = \frac{0}{0}$

$\lim_{x\to 0} \frac{tg\,x}{x} = \lim_{x\to 0} \frac{\frac{sen\,x}{cos\,x}}{\frac{x}{1}} = \lim_{x\to 0} \frac{\frac{sen\,x}{x}}{\cancel{cos\,x}} = \frac{1}{cos\,x} = \lim_{x\to 0} \frac{1}{cos\,x}$

$= \frac{1}{cos(0)} = \frac{1}{1} = \boxed{1}$

b) $\lim_{x\to 0} \frac{sen\,kx}{x} = K = \frac{sen(0)K}{0} = \frac{0}{0}$

$\lim_{x\to 0} \frac{sen\,kx}{x} = \lim_{x\to 0} \frac{K\,sen\,kx}{Kx} = 1 = \boxed{\lim_{x\to 0} K = K}$

c) $\lim_{x\to 0} \frac{1-cos\,x}{x^2} = \frac{1}{2} = \frac{1-cos(0)}{(0)} = \frac{1-1}{0} = \frac{0}{0}$

$\lim_{x\to 0} \frac{1-cos\,x}{x^2} \cdot \frac{1+cos\,x}{1+cos\,x} = \lim_{x\to 0} \frac{1^2-(cos\,x)^2}{x^2(1+cos\,x)} =$

$\lim_{x\to 0} \frac{1-cos^2 x}{x^2(1+cos\,x)} = \lim_{x\to 0} \frac{sen^2 x}{x^2(1+cos\,x)} = \lim_{x\to 0} \frac{sen\,x}{x} \cdot \frac{sen\,x}{x} \cdot \frac{1}{1+cos\,x}$

$\lim_{x\to 0} \frac{1}{1+cos\,x} = \frac{1}{1+cos(0)} = \frac{1}{1+1} = \boxed{\frac{1}{2}}$

Escaneado con CamScanner

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d) $\lim_{x \to 0} \frac{1-\cos x}{x} = 0 = \frac{1-\cos(0)}{(0)} = \frac{1-1}{0} = \frac{0}{0}$

$\lim_{x \to 0} \frac{1-\cos x}{x} \cdot \frac{1+\cos x}{1+\cos x} = \frac{1^2-(\cos x)^2}{x(1+\cos x)} =$

$\lim_{x \to 0} \frac{1-\cos^2 x}{x(1+\cos x)}$ $\lim_{x \to 0} \frac{\operatorname{sen}^2 x}{x(1+\cos x)} = \frac{\operatorname{sen} x}{\cancel{x}} \cdot \frac{\operatorname{sen} x}{1+\cos x}$

$\lim_{x \to 0} \frac{\operatorname{sen} x}{1+\cos x} = \frac{\operatorname{sen}(0)}{1+\cos(0)} = \frac{0}{1+1} = \frac{0}{2} = 0$

e) $\lim_{x \to 0} \frac{\operatorname{sen} x^2}{x} = 0 = \frac{\operatorname{sen}(0)^2}{(0)} = \frac{0}{0}$

$\lim_{x \to 0} \frac{\operatorname{sen} x^2}{x} = \lim_{x \to 0} \left(\frac{\operatorname{sen}(x)(\operatorname{sen} x)}{x} \right) \lim_{x \to 0} \left(\operatorname{sen} x \left(\frac{\operatorname{sen}(x)}{x} \right) \right)$

$= \operatorname{sen}(0)(1) = 0$